

Development of a Digital Twin-Based Predictive Maintenance System for Industrial Motor

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Problem and Objective

- Traditional maintenance is reactive or scheduled; lacks real-time diagnostics.
- Induction motor failures cause unplanned downtime and high operational costs.
- Solving this leads to extended motor life, reduced failures, and energy savings.
- **Project Objectives:**
 - Design a predictive maintenance system using digital twin technology.
 - Achieve high fault detection accuracy through real-time sensor analytics.
 - Integrate ML models with live motor data using ESP32 and MQTT.
 - Visualize motor condition using a UI based interface.
 - Ensure scalability for other industrial equipment and sensor types.

Problem Definition

This project focuses on building a smart predictive maintenance system that continuously monitors industrial motor health using real-time sensor data. It aims to reduce unplanned downtimes by predicting faults early, ensuring a scalable, cost-effective, and autonomous solution fit for modern IIoT environments.

Problem Solution

- A predictive maintenance system using a digital twin that mirrors real-time motor data.
- Sensor data (temperature, vibration) is captured via ESP32 and transmitted using MQTT.
- A lightweight ML model analyzes the data to predict faults before they occur.
- A Unity3D-based 3D digital twin visualizes the motor's live condition and predicted anomalies.
- Enables proactive decision-making, reduces downtime, and extends motor lifespan.

Literature Review and Survey

● Key Research Papers Studied:

- Interface Development for Digital Twin of an Electric Motor (IEEE Access, 2022)
- Explainable Data-Driven Digital Twins for Predicting Battery States (IEEE Access, 2024)
- Digital Twin Implementation for Additive Manufacturing Robotic Cell (IEEE LATAM, 2024)
- A Survey on Digital Twins: Enabling Technologies and Use Cases (IEEE Sensors, 2025)
- Digital Twins for IoT-Driven Energy Systems: A Survey (IEEE Access, 2024)

● Tools/Technologies Referenced:

- MQTT communication, ROS-based control, Unity3D visualization
- ML models (LSTM, Decision Trees), Sensor fusion, Real-time dashboards

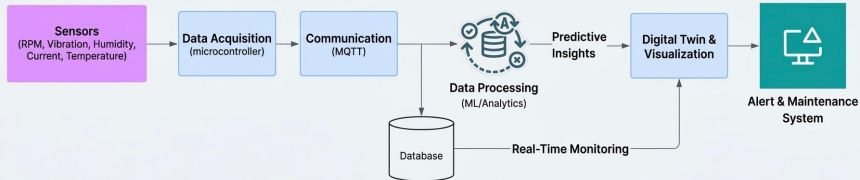
● Insights and Gaps Identified:

- Most works lack immersive AR visualization and focus on static data
- Limited use of lightweight ML models suitable for edge devices
- Few implementations use real-time predictive feedback on physical components

● How Our Project Builds Upon Them:

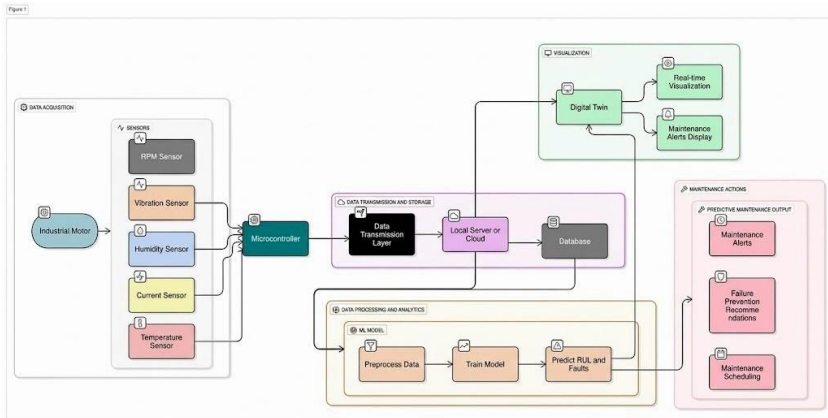
- Combines live sensor fusion with ML-based fault prediction
- Adds 3D/AR-based visualization of digital twin using Unity
- Targets edge-based, scalable deployment using ESP32 + MQTT

Block Diagram



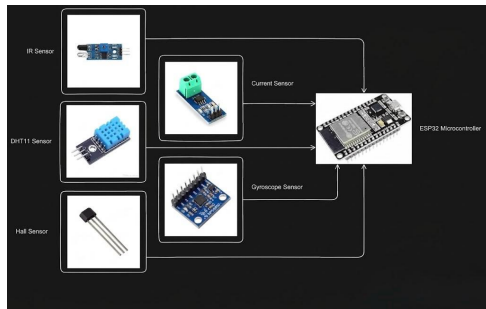
Block Diagram of the Proposed System

System Architecture



Hardware Components Used

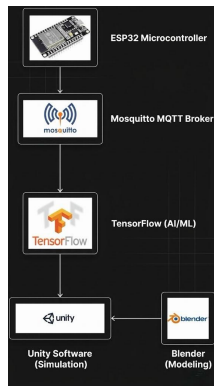
- Sensors: RPM, Torque, Current, Voltage
- ESP32 / Raspberry Pi (if applicable)
- Wiring, power supply, relays
- Breadboard and connectors



Sensor-Microcontroller layout

Software Tools and Frameworks

- Unity3D – 3D Visualization
- ROS (Robot Operating System)
- Python / C++ ROS Nodes
- MQTT Broker (Local/Remote)
- Data Collection & Empirical Modeling Tools



From sensing to visualization

Project Abstract

- A smart predictive maintenance system using a Digital Twin of an induction motor.
- Real-time sensor data (temperature, vibration, current) is collected via ESP32 and transmitted using MQTT.
- A lightweight ML model predicts motor health, potential faults, and degradation patterns.
- Unity3D visualizes the 3D twin and reflects predicted issues with visual cues and alerts.
- System supports optional AR mode for immersive, interactive fault diagnosis and remote visibility.

Temperature Sensor Readings from Motor

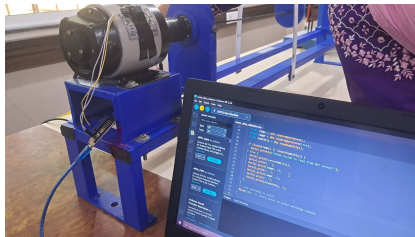
Objective: Capture real-world thermal behavior of the motor for Digital Twin validation.

Setup:

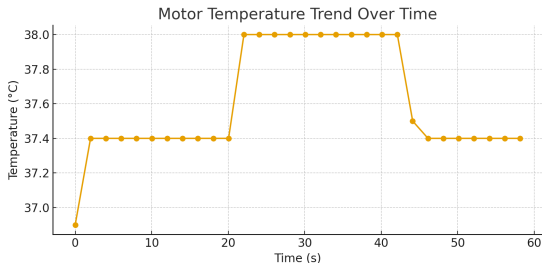
- **Sensor:** DHT11 attached near motor casing.
- **Interface:** ESP32 for data acquisition.
- **Data Handling:** Temperature readings collected via serial monitor for analysis.

Observation:

- Recorded temperature variations under different load conditions.
- Data will be used in the next phase to align digital model response with physical system.



Temperature Trend of Motor



Plot Information:

- Data collected from DHT11 sensor mounted on motor casing.
- Time axis derived from millisecond timestamps (converted to seconds).
- Data collected for 200,400 and 600 rpm.
- Temperature in °C plotted over time, ranging from **36.9 °C to 58 °C**.
- Humidity remained consistently high at around **99%**.

Predictive Maintenance — Kaggle Dataset & RF Evaluation

Random Forest on Kaggle Dataset (Validation)

- Dataset: 10,000 samples, 14 features — air temp, process temp, torque, tool wear, rotation speed
- Split: 80% training / 20% testing with stratified sampling
- Result:
 - 98% overall accuracy
 - Precision: 0.98, Recall: 1.00 (Normal class)
- Limitation: Recall 0.49 for fault class due to class imbalance (real motors rarely fail)
- Purpose: Validates RF approach on benchmark data before applying to real sensor data

	Class 0 = Normal	—	Class 1 = Fault	
	precision		recall	f1-score
0	0.98		1.00	0.99
1	0.86		0.49	0.62
accuracy				0.98
macro avg	0.92		0.74	0.81
weighted avg	0.98		0.98	0.98

Why additional models?

While Random Forest achieves high accuracy, it cannot detect gradual degradation within the same operating state. Hence, Isolation Forest and Autoencoder models are used, trained on healthy baseline data, to continuously monitor motor health and detect anomalies before failure.

Note: Lower fault recall is expected - real machines fail rarely, causing natural class imbalance.

Tri-Layer Predictive Maintenance Architecture

Multi-Stage Decision Logic

- 1 **Isolation Forest (Unsupervised):** Trained on 454 healthy windows. Flags anomalies in $\sim 1\%$ of data (noise/faults).
- 2 **Rule-Based Layer (Physics):** Overrides ML for high-confidence signatures (e.g., Temp $> 68^{\circ}\text{C}$). Adds deterministic reliability.
- 3 **XGBoost (Supervised):** Classifies 5 fault types on 7,750 rows. 99% test accuracy.

Health & RUL Metrics

- **Health Score (0-100):** Calculated via threshold penalties (Vib, Temp, Kurtosis).
- **RUL Estimation:** Linear regression on health slope; predicts hours to critical failure.

Model Comparison & Performance

Layer	Metric	Function/Result
Iso Forest	300 Trees	Baseline health boundary
Rule Layer	Physics-Ref	92% Conf. (Overheating)
XGBoost	99% Acc.	Final classification layer
RUL	Regressor	Predicts hours to limit

Validation (Real vs. Synthetic)

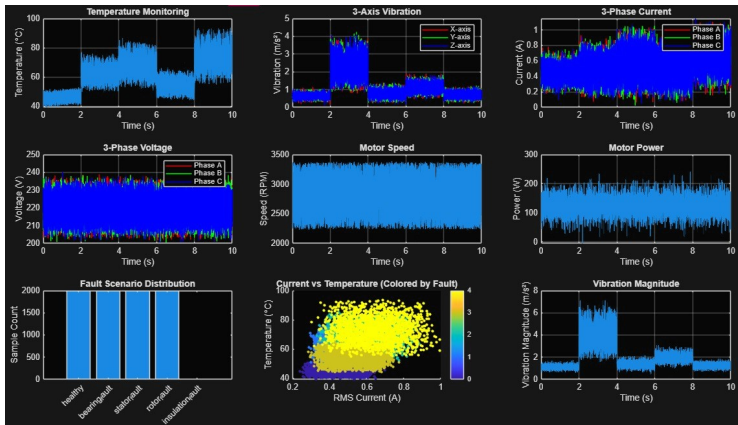
- **Real (200/400 RPM):** Validated as Healthy (88-100% Health).
- **Fault Scenarios:**
 - **Bearing:** Caught at Window 41 (RUL 1.5h).
 - **Overheat:** 92% Conf. via Rule Layer.
 - **Stator/Rotor:** Correct tracking of transitions.

RAMAN Motor Digital Twin – Development Workflow

Key Activities:

- **Baseline Model Setup:** Used MATLAB's ACIM FOC (QEP) example with corrections for per-unit inconsistencies (N_{base} , f_{base} , T_{base}) and customization for the RAMAN Motor.
- **Synthetic Data Framework:** Defined healthy and fault scenarios generating electrical, thermal, vibration, and mechanical signals. Designed a nine-panel dashboard for visualization.
- **Feature Engineering:** Extracted features such as RMS, THD, and power factor, and prepared a labeled dataset (`raman_motor_dataset.csv`) for machine learning.
- **Simulink Integration:** Developed an end-to-end pipeline with monitoring scopes, solver, and logging. Execution blocked by unresolved `PU_System.N_base` dependency and hardware configuration.
- **Final Outcome:** The final outcome is a hybrid Physics-Based and Data-Driven digital twin. Synthetic datasets are ready for ML training, while the Simulink framework is in place but requires resolution of parameter and hardware issues for full execution.

MATLAB Simulation Results



Simulation output visualizing motor behavior using MATLAB.

GUI & Real-Time Digital Twin Dashboard

Interactive Digital Twin (Unity)

- **60 FPS 3D Render:** Real-time orientation and state visualization.
- **X-Ray Mode:** Transparent housing to inspect internal components (Stator, Rotor, Bearings).

Telemetry & Analytics

- **Live Sensor Feed:** Vibration magnitude, 3-axis acceleration, and temperature.
- **Predictive Logic:** Health Score (0-100), RUL (Hours), and Maintenance Trends.

Intelligent Alerting

- **Visual Feedback:** Components glow red/orange during fault states.
- **Action Logs:** Auto-generated maintenance tasks (e.g., "Schedule lubrication").

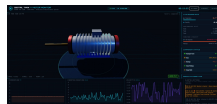
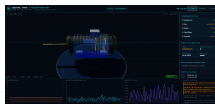
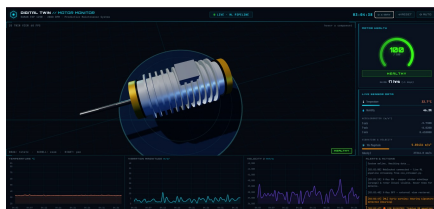


Figure: Clockwise from top: Main Monitoring, Fault Detection, and X-Ray Internal Inspection.

Issues Faced and Resolutions

During the implementation, several practical challenges were encountered and systematically resolved:

Issue	Cause	Resolution
Inaccurate Temperature Readings	DHT11 sensor not in direct thermal contact with motor surface	Applied thermal grease between motor body and sensor to maintain constant conductivity.
Fluctuating Humidity Values	DHT11 sensitivity to motor heat and environmental interference	Implemented moving average filter on ESP32 for stable values.
Vibration Data Instability	MPU6050 produced noisy signals during high RPM	Used Kalman filtering in preprocessing to smooth out vibration readings.
Integration Delay with MATLAB	Serial communication lag from ESP32	Optimized baud rate to 115200 and used MATLAB's <code>serialport()</code> for efficient data handling.

Outcome & Future Improvements

Outcome:

- Achieved **98% accuracy** on the predictive maintenance dataset, along with effective anomaly detection and health monitoring.
- Successfully demonstrated a **real-time Digital Twin system** with an interactive GUI for monitoring motor condition.
- Implemented and validated a **multi-layer predictive maintenance pipeline** (Random Forest, Isolation Forest, XGBoost) for fault detection and classification.
- Integrated **sensor data, machine learning, and visualization** into a unified framework enabling live condition tracking, health score estimation, and predictive insights.

Future Improvements:

- Upgrade sensors for higher precision (replace DHT11 with DHT22 / industrial-grade sensors).
- Incorporate advanced ML models (e.g., LSTM, Gradient Boosting) for better fault prediction.
- Improve scalability for real-time deployment in industrial environments.
- Enhance the system with **cloud-based data storage and remote monitoring capabilities**.

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